

MATH 708. SPRING 2026. HOMEWORK 3. DUE ON FEBRUARY 27.

- Unless otherwise stated, the exercises are from Voisin's textbook.
- The homework problems may be submitted orally (during assigned office hours), in written/typed form, or as a combination of the two.
- You may submit any subset of problems totaling up to 100 points.
- If you submit more than 100 points worth, I will grade only the first 100 points you designate.
- Do not write anything on the cover sheet, as it will be used for grading. Bring it to office hours when you submit problems orally, and staple it to your written/typed solutions.

Problem 1. (10 points) Let M be a complex submanifold of $\mathbb{P}^3$ given by

$$x_0^2 + x_1^2 + x_2^2 + x_3^2 = 0.$$

Let I be the corresponding almost complex structure on M viewed as a smooth manifold. Show that (M, I) and $(M, -I)$ define isomorphic complex manifolds.

Problem 2. (15 points) Let X be an almost complex manifold. For every form α of type (p, q) on X , define $\partial\alpha$ (resp. $\bar{\partial}\alpha$) to be the projection of $d\alpha$ onto $A_X^{p+1, q}$ (resp. $A_X^{p, q+1}$). Extend ∂ and $\bar{\partial}$ by linearity to maps $A^*(X) \rightarrow A^*(X)$. Then the following conditions are equivalent:

- (1) X is a complex manifold;
- (2) $d\alpha = \partial\alpha + \bar{\partial}\alpha$ for all $\alpha \in A^*(X)$;
- (3) for $\alpha \in A_X^{1,0}$, the projection of $d\alpha$ to $A_X^{0,2}$ is zero;
- (4) $\bar{\partial}^2 = 0$.

Problem 3. (5 points) Let V be a complex vector space, and let $J : V_{\mathbb{R}} \rightarrow V_{\mathbb{R}}$ denote the complex structure, i.e. $J(u) = iu$ when we view V as a real vector space. Construct a bijection between positive definite Hermitian forms h on V , and positive definite real inner products g on $V_{\mathbb{R}}$ compatible with J (which means that $g(Ju, Jv) = g(u, v)$, i.e. J is an isometry).

Problem 4. (15 points) Let $(V, (\cdot, \cdot))$ be a four-dimensional Euclidean vector space. Show that the set of all compatible constant almost complex structures on V consists of two copies of the 2-sphere.

Problem 5. (10 points) Let $(V, (\cdot, \cdot))$ be a Euclidean vector space, and let I , J , and K be compatible almost complex structures with

$$K = I \circ J = -J \circ I.$$

Show that V becomes in a natural way a vector space over the quaternions. The associated skew-symmetric forms are denoted by ω_I , ω_J , and ω_K . Show that $\omega_J + i\omega_K$ is a form of type $(2, 0)$ with respect to I .

Problem 6. (15 points) Exercise 1, page 82

Problem 7. (10 points) Let

$$\omega = \frac{i}{2\pi} \sum_{j=1}^n dz_j \wedge d\bar{z}_j$$

be the standard Kähler form on $\mathbb{C}^n$. Show that one can write $\omega = i \partial \bar{\partial} \varphi$ for some positive function φ , and determine φ .

Problem 8. (5 points) Show that the exterior product induces a multiplication on the full Dolbeault cohomology $\bigoplus_{p,q} H^{p,q}(X)$ of a complex manifold X , and that this yields a $\mathbb{Z}^2$ -graded super-commutative algebra.

Problem 9. (10 points) The set of all Kähler forms on a compact complex manifold X is an open convex cone in the linear space

$$\{\omega \in A^{1,1}(X) \cap A^2(X) \mid d\omega = 0\}.$$

Problem 10. Let $\mathbb{D} = \{\|z\| < 1\} \subset \mathbb{C}^n$ be the unit ball.

(5 points) Show that the form $\omega = i \partial \bar{\partial} \log(1 - \|z\|^2)$ is Kähler on $\mathbb{D}$.

(5 points) Identify $\mathbb{D}$ with the open subset of $\mathbb{P}^n$ consisting of points $[v]$ with $(v, v) > 0$, where $(\cdot, \cdot)$ is the (non-positive) Hermitian product on $\mathbb{C}^{n+1}$ given by the diagonal matrix $\text{diag}(1, -1, \dots, -1)$. Conclude that the group $SU(1, n)$ acts on $\mathbb{D}$ and preserves ω .

(5 points) Show that every ball quotient $\mathbb{D}/\Gamma$ is a Kähler manifold. Here $\Gamma \subset SU(1, n)$ is a discrete subgroup acting freely on $\mathbb{D}$.

Problem 11. (15 points) Show $\int_{\mathbb{P}^n} \omega^n = 1$, where ω is the Fubini–Study form.

Problem 12. (5 points) Let X and Y be two Kähler manifolds. Show that the product $X \times Y$ admits a natural Kähler structure, too.

Problem 13. (5 points) Let L be a holomorphic line bundle on a complex manifold X , and let $s_1, \dots, s_k$ be global holomorphic sections of L generating L everywhere, i.e. for every $x \in X$ at least one $s_i(x)$ is nonzero. Show that one can define a Hermitian metric on L by the formula

$$h(t) = \frac{|\psi(t)|^2}{\sum |\psi(s_i(x))|^2},$$

where $t \in L_x$ is a point in the fibre, and ψ is a local trivialization of L .

Problem 14. (15 points) Exercise 2, page 82

Problem 15. (10 points) Let ∇ be a connection on a vector bundle E . Describe the induced connection on the line bundle $\det(E) = \bigwedge^{\text{rk}(E)} E$. What is its connection matrix? What is its curvature?